

◆ Normalization (Min-Max Scaling)

Range: $[0, 1]$ (ya kisi aur specified range mein)

◆ Use Normalization when:

- Tumhara data bounded range mein hona chahiye (jaise 0 se 1).
 - Distance-based algorithms use kar rahe ho, jese:
 - K-Nearest Neighbors (KNN)
 - K-Means Clustering
 - Neural Networks (especially when using activation functions like sigmoid/tanh)
 - Jab tumhara data not normally distributed ho.
 - Jab outliers ka effect kam ho ya outliers ko handle kiya gaya ho.
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◆ Standardization (Z-score Scaling)

Range: Mean = 0, Std = 1 (but no fixed bounds like $[0,1]$)

◆ Use Standardization when:

- Tumhara data normal distribution ya approximately normal distribution follow karta hai.
- Algorithms jahan assumption hoti hai ke data Gaussian ho, jaise:
 - Linear Regression
 - Logistic Regression
 - Support Vector Machines (SVM)
 - Principal Component Analysis (PCA)
- Jab outliers present hain, kyunki standardization outliers ko zyada handle kar pata hai compared to normalization.